

SESSION-9  
MODULE-5

```
-- =====  
-- Session 9 — Transactions & Error Handling  
-- COMMIT, ROLLBACK, SAVEPOINT, Triggers  
-- Run block by block in phpMyAdmin / MySQL Workbench  
--  
-- IMPORTANT: transactions only work on InnoDB tables. MySQL's older  
-- MyISAM engine ignores COMMIT/ROLLBACK entirely and just saves  
-- everything immediately. All tables below explicitly use InnoDB  
-- (it's also phpMyAdmin/MySQL's default nowadays, but stating it  
-- removes any doubt).  
-- =====
```

USE instaclone;

DROP TRIGGER IF EXISTS trg\_wallet\_deduct;

DROP TABLE IF EXISTS Cart;

DROP TABLE IF EXISTS OrderItems;

DROP TABLE IF EXISTS Orders;

DROP TABLE IF EXISTS Wallet;

-- ----- Task 1: insert + COMMIT, verify it persists -----

```
CREATE TABLE Orders (  
    order_id INT AUTO_INCREMENT PRIMARY KEY,  
    customer VARCHAR(100),  
    order_total DECIMAL(10,2)  
) ENGINE=InnoDB;
```

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START TRANSACTION;

INSERT INTO Orders (customer, order\_total) VALUES ('Priya Mehta', 850.00);

COMMIT; -- makes the insert permanent — survives a disconnect/reconnect

-- Verify: close this tab / reconnect to the server, then run:

SELECT \* FROM Orders;

-- The row should still be there, because COMMIT already wrote it to disk.

-- ----- Task 2: insert two items, then ROLLBACK -----

```
CREATE TABLE OrderItems (  
    item_id INT AUTO_INCREMENT PRIMARY KEY,  
    order_id INT,  
    item_name VARCHAR(100),  
    price DECIMAL(10,2)  
) ENGINE=InnoDB;
```

START TRANSACTION;

INSERT INTO OrderItems (order\_id, item\_name, price) VALUES (1, 'Paneer Butter Masala', 320.00);

INSERT INTO OrderItems (order\_id, item\_name, price) VALUES (1, 'Garlic Naan', 60.00);

-- Changing our mind before committing:

ROLLBACK; -- undoes BOTH inserts above, as if they never happened

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```
SELECT * FROM OrderItems;
```

```
-- Expected: empty table — neither item was actually saved.
```

```
-- ----- Task 3: SAVEPOINT — undo only the last addition -----
```

```
CREATE TABLE Cart (  
    cart_id INT AUTO_INCREMENT PRIMARY KEY,  
    product VARCHAR(100)  
) ENGINE=InnoDB;
```

```
START TRANSACTION;
```

```
INSERT INTO Cart (product) VALUES ('Wireless Mouse');
```

```
SAVEPOINT after_first_product; -- bookmark right here
```

```
INSERT INTO Cart (product) VALUES ('USB-C Cable');
```

```
-- Changed our mind about the cable only — not the mouse:
```

```
ROLLBACK TO SAVEPOINT after_first_product;
```

```
COMMIT; -- now finalize whatever state remains after the partial rollback
```

```
SELECT * FROM Cart;
```

```
-- Expected: only 'Wireless Mouse' — the cable insert was undone,
```

```
-- but everything before the savepoint survives.
```

```
-- ----- Task 4: trigger to auto-deduct from Wallet -----
```

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```
CREATE TABLE Wallet (  
    user_id INT PRIMARY KEY,  
    balance DECIMAL(10,2)  
) ENGINE=InnoDB;
```

```
INSERT INTO Wallet (user_id, balance) VALUES  
  
    (1, 5000.00),  
  
    (2, 2000.00);
```

```
-- WalletTransactions logs each purchase; the trigger below fires  
-- automatically after each insert and adjusts the matching balance.
```

```
DROP TABLE IF EXISTS WalletTransactions;  
  
CREATE TABLE WalletTransactions (  
    txn_id INT AUTO_INCREMENT PRIMARY KEY,  
    user_id INT,  
    amount DECIMAL(10,2) -- the purchase amount to deduct  
) ENGINE=InnoDB;
```

```
DELIMITER //  
  
CREATE TRIGGER trg_wallet_deduct  
AFTER INSERT ON WalletTransactions  
FOR EACH ROW  
BEGIN  
  
    UPDATE Wallet  
  
    SET balance = balance - NEW.amount  
  
    WHERE user_id = NEW.user_id;  
  
END //
```

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DELIMITER ;

-- Test it:

SELECT \* FROM Wallet; -- before: 5000.00, 2000.00

INSERT INTO WalletTransactions (user\_id, amount) VALUES (1, 750.00);

SELECT \* FROM Wallet; -- after: user 1 -> 4250.00

-- ----- Task 5: buggy trigger scenario + fix -----

-- (Written directly here rather than pasted from ChatGPT — the

-- deliverable the task actually wants is the buggy example, the test

-- that exposes it, the fixed version, and an explanation of the fix.)

-- --- The buggy version ---

-- A common real mistake: writing the deduction backwards, i.e. ADDING

-- the amount to the balance instead of subtracting it.

-- DROP TRIGGER IF EXISTS trg\_wallet\_deduct\_buggy;

-- DELIMITER //

-- CREATE TRIGGER trg\_wallet\_deduct\_buggy

-- AFTER INSERT ON WalletTransactions

-- FOR EACH ROW

-- BEGIN

-- UPDATE Wallet

-- SET balance = balance + NEW.amount -- BUG: should be minus, not plus

-- WHERE user\_id = NEW.user\_id;

-- END //

-- DELIMITER ;

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-- --- The test that exposes the bug ---

-- SELECT balance FROM Wallet WHERE user\_id = 2;      -- 2000.00

-- INSERT INTO WalletTransactions (user\_id, amount) VALUES (2, 300.00);

-- SELECT balance FROM Wallet WHERE user\_id = 2;      -- BUGGY result: 2300.00

-- A 300 purchase should LOWER the balance to 1700.00, not raise it to

-- 2300.00 — the test makes the sign error obvious immediately.

-- --- The fix ---

-- Simply flip + back to -, which is exactly what trg\_wallet\_deduct

-- above already does correctly. Re-running the same test against the

-- correct trigger:

SELECT balance FROM Wallet WHERE user\_id = 2;      -- 2000.00

INSERT INTO WalletTransactions (user\_id, amount) VALUES (2, 300.00);

SELECT balance FROM Wallet WHERE user\_id = 2;      -- correct: 1700.00

-- Explanation: the bug is a one-character sign error (+ instead of -)

-- that's easy to miss on read-through because the trigger still runs

-- without any error — it just silently moves money the wrong

-- direction. This is why Task 5 pairs a trigger with an explicit

-- before/after balance check: a trigger with no test can look

-- completely correct and still be financially wrong.

## SESSION-9

### MODULE-5

The image displays two screenshots of the phpMyAdmin web interface, showing SQL query results for two different tables.

**Top Screenshot: Table 'orderItems'**

- URL: `localhost/phpmyadmin/index.php?route=/sql&pos=0&db=instadone&table=orderItems`
- Query: `SELECT * FROM `orderItems``
- Result: MySQL returned an empty result set (i.e. zero rows). (Query took 0.0005 seconds.)
- Table structure: `item_id, order_id, item_name, price`

**Bottom Screenshot: Table 'wallet'**

- URL: `localhost/phpmyadmin/index.php?route=/sql&pos=0&db=instadone&table=wallet`
- Query: `SELECT * FROM `wallet``
- Result: Showing rows 0 - 1 (2 total). Query took 0.0009 seconds.
- Table structure: `user_id, balance`
- Data:

| user_id | balance |
|---------|---------|
| 1       | 4250.00 |
| 2       | 1700.00 |

## SESSION-9

### MODULE-5

localhost / 127.0.0.1 / instadone / x New tab

localhost/phpmyadmin/index.php?route=/sql&pos=0&db=instadone&table=wallettransactions

phpMyAdmin

Recent Favorites

task5\_safe\_divide

Tables

- New
- cart
- foodorder
- influencers
- movies
- orderitems
- orders
- payments
- playlist
- playlists
- posts
- products
- restaurant
- restaurants
- restaurant\_ratings
- spotifyuser
- users
- wallet
- wallettransactions

Views

- mydb
- mysql
- performance\_schema

Showing rows 0 - 1 (2 total, Query took 0.0006 seconds)

SELECT \* FROM `wallettransactions`

Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]

Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

|                          | txn_id | user_id | amount |
|--------------------------|--------|---------|--------|
| <input type="checkbox"/> | 1      | 1       | 750.00 |
| <input type="checkbox"/> | 2      | 2       | 300.00 |

Check all With selected: Edit Copy Delete Export

Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

Bookmark this SQL query

Label:  ☐ Let every user access this bookmark

Bookmark this SQL query

localhost/phpmyadmin/index.php?route=/sql&pos=0&db=instadone&table=wallettransactions

32°C Mostly clear

Search

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